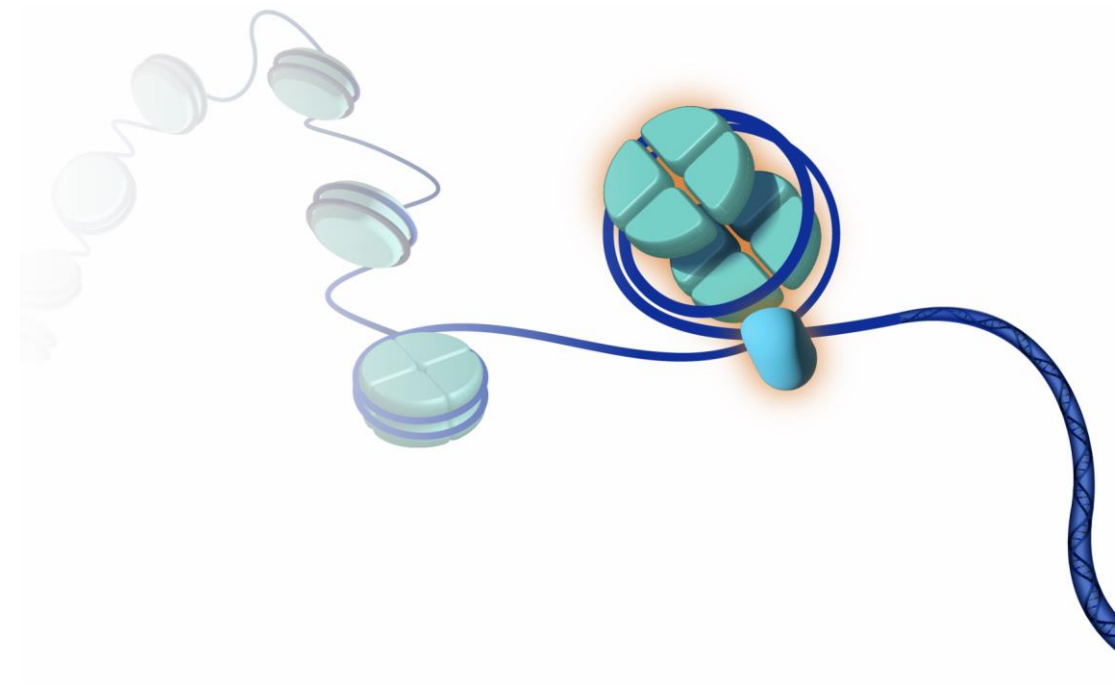
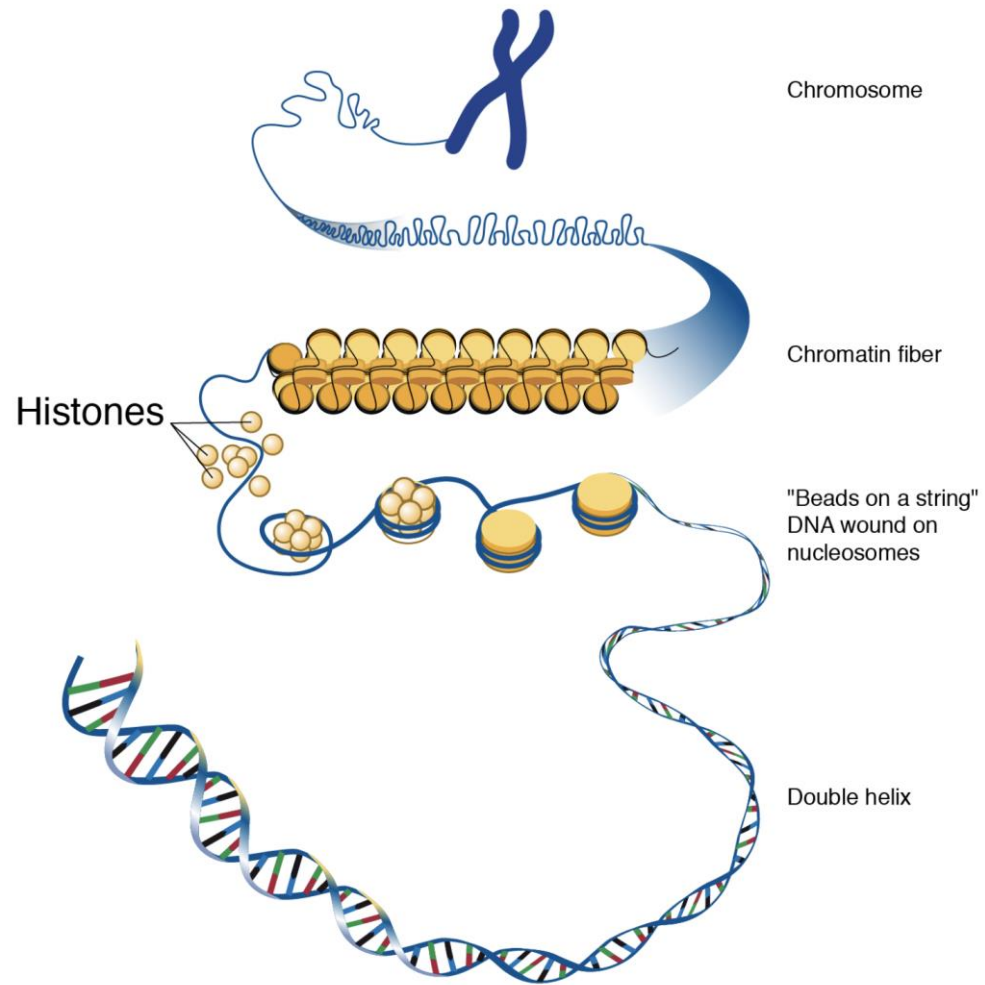


ChIP-seq Analyses

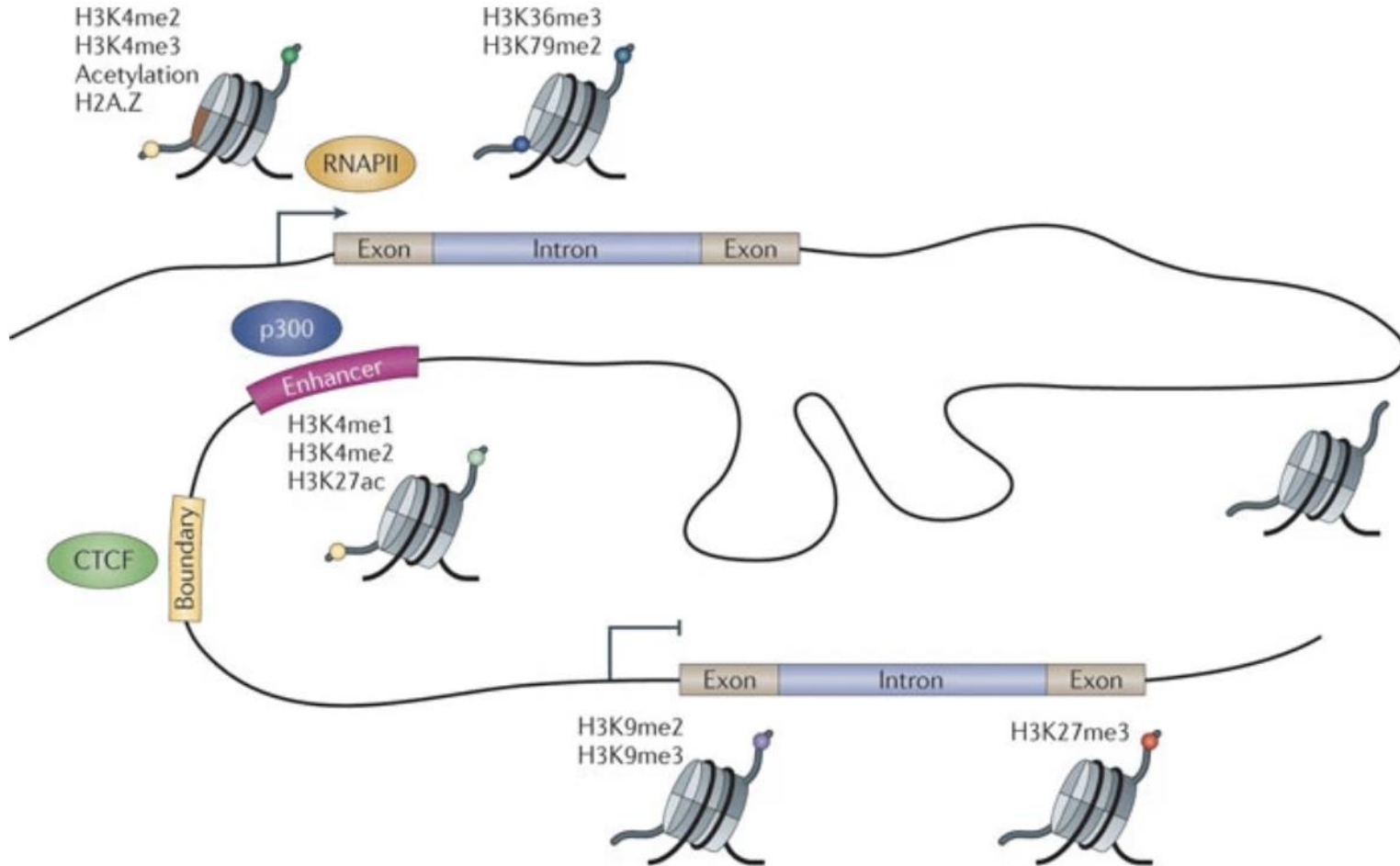
Dania Machlab

October 7, 2026

Chromatin organization



Histone marks



- **Active promoters:**

- H3K4me2,
- H3K4me3
- acetylation (ac)
- H2A.Z

- **Transcribed regions:**

- H3K36me3
- H3K79me2

- **Repressed genes:**

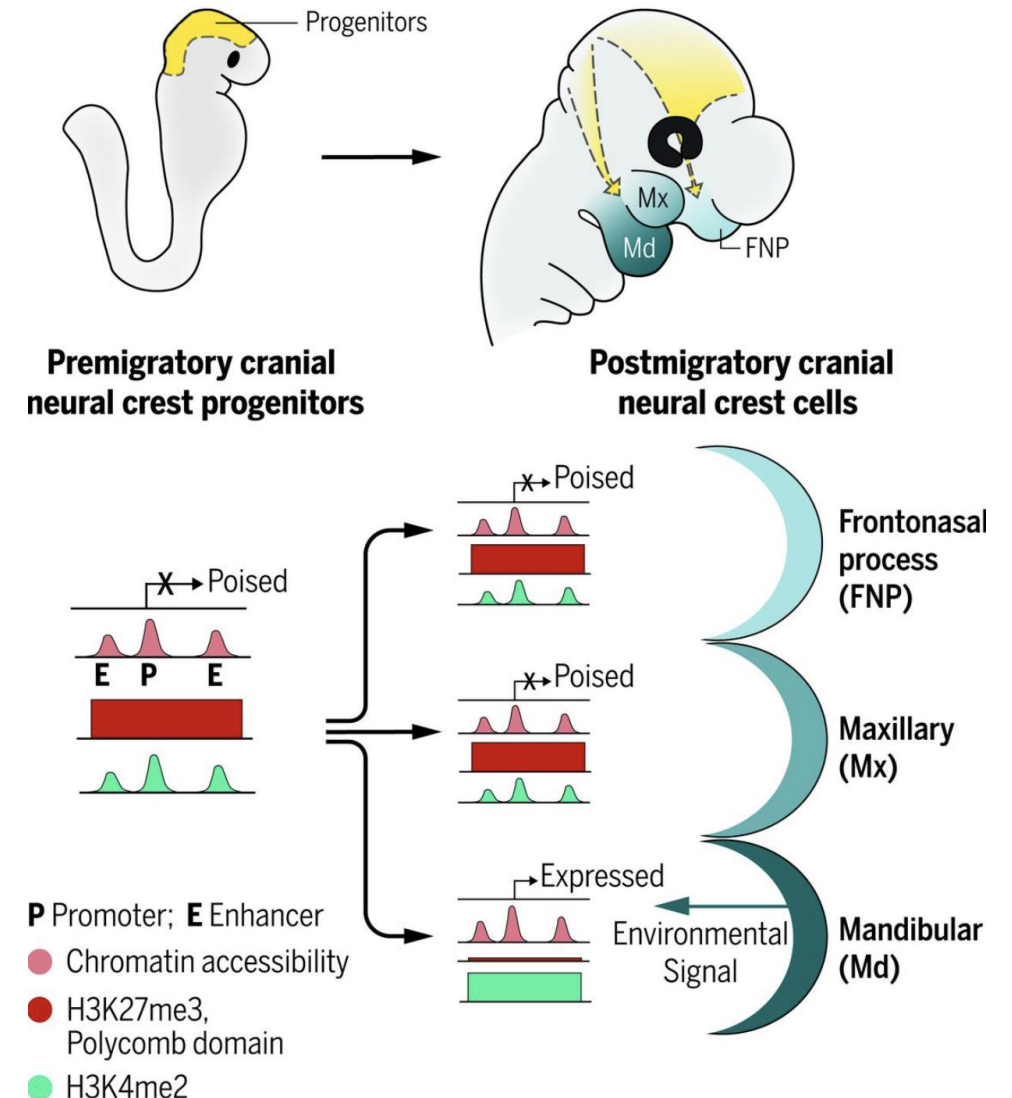
- H3K9me2
- H3K9me3
- H3K27me3

- **Enhancers:**

- H3K4me1
- H3K4me2
- H3K27ac

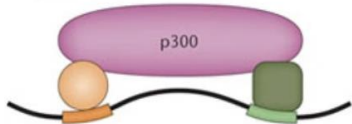
The epigenetics of face-making

- How is it that our earlobes are attached to our ears and not our chins?
- Bivalent promoters and enhancers in early stages: H3K27Me3+ and H3K4Me2+
- Cranial NC cell plasticity is maintained through migration until post-migratory stages
- When the migrating neurons find the final location different developmental programs are unlocked

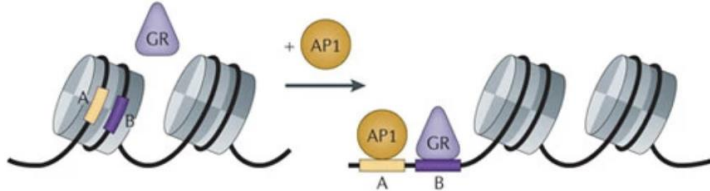


DNA binding proteins – transcription factors

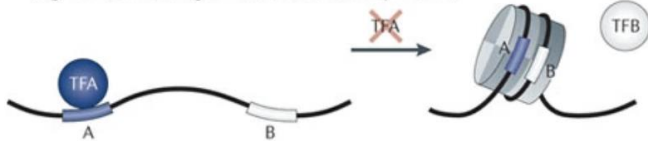
a Co-binding to common cofactors or common complexes (transcriptional synergy)



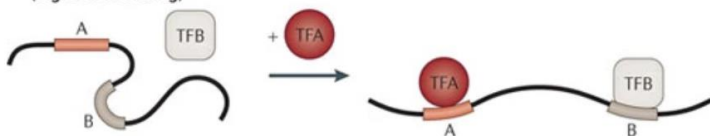
b Activating chromatin remodelling (e.g. a pioneer TF)



c Blocking nucleosome repositioning (e.g. 'assisted loading' or 'collaborative competition')

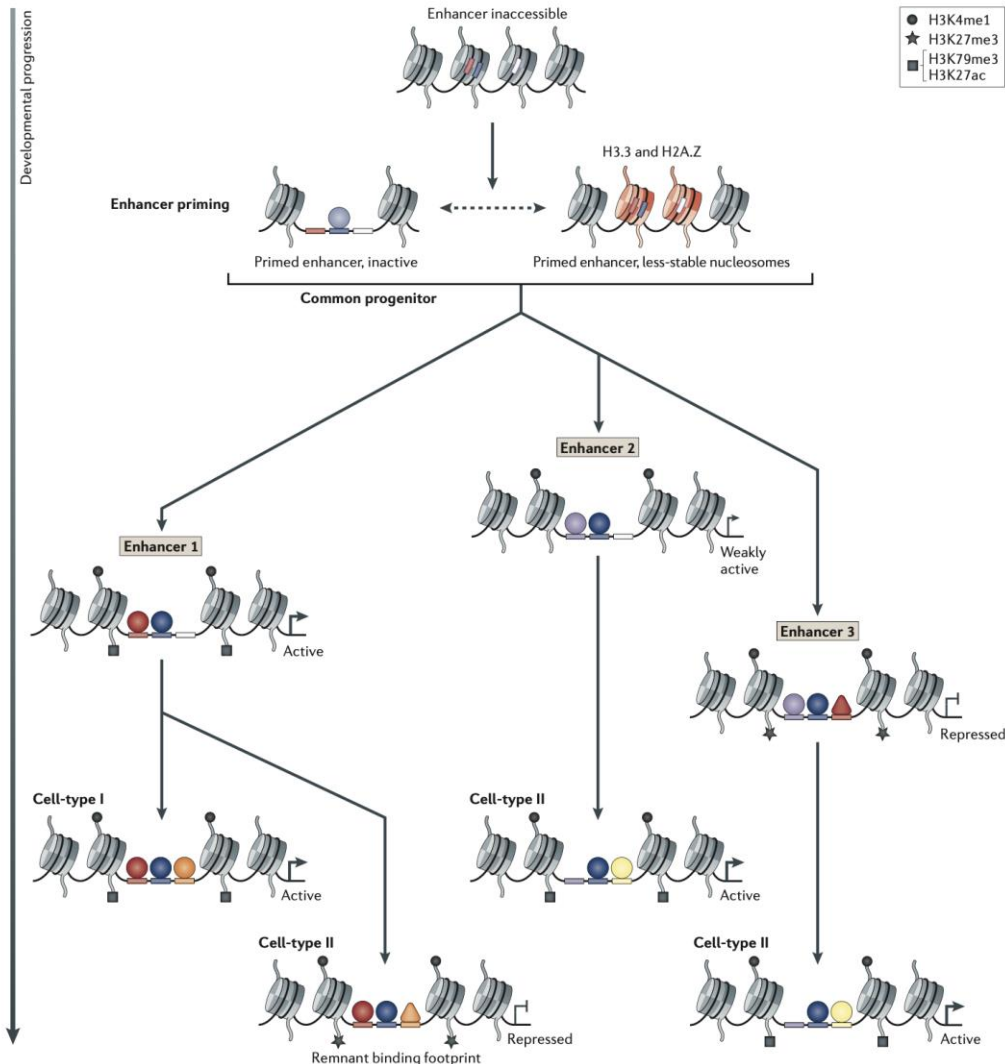


d Architectural role (e.g. DNA bending)



- TFs bound to the same enhancer may recruit a common cofactor (e.g. p300)
- Chromatin re-modelling can expose new TF binding sites
- Bound TFs can also prevent nucleosome repositioning, allowing other TFs to bind to exposed sites
- Some TFs can induce local DNA bending which can increase the affinity of other TFs

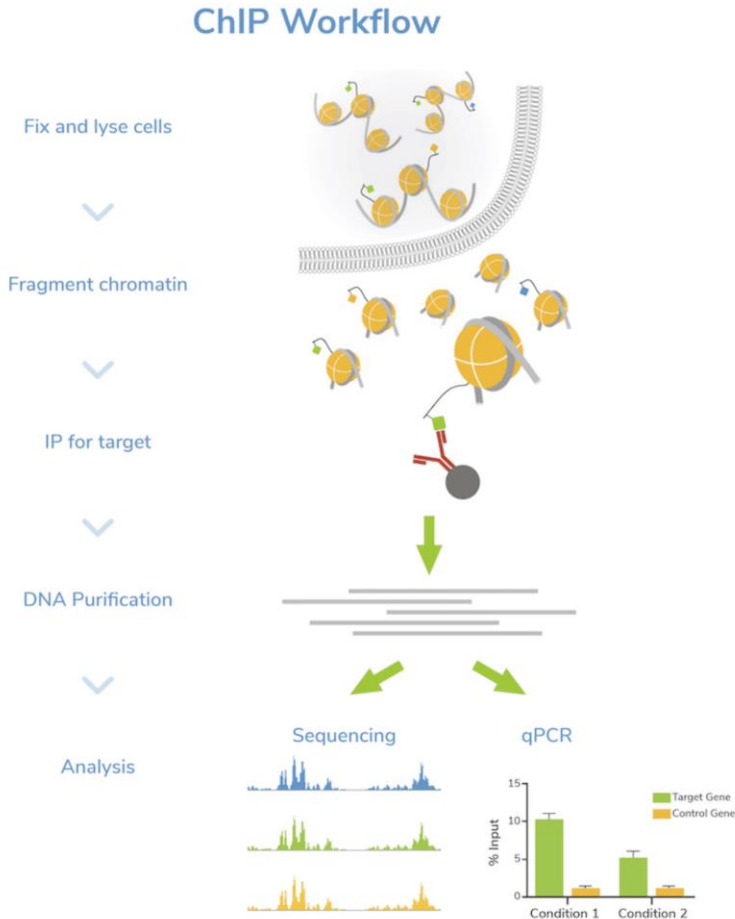
Why should we care about chromatin modifications/players?



- TFs play key roles in regulating gene expression on a genome-wide level.
- Histone marks and their combinations also reflect gene activity
- Changes over development
- Assay which captures interesting global regulation on the chromatin level which can drive and influence what we see on the gene level
- There is still a lot we don't understand



ChIP-seq: Chromatin Immunoprecipitation followed by sequencing



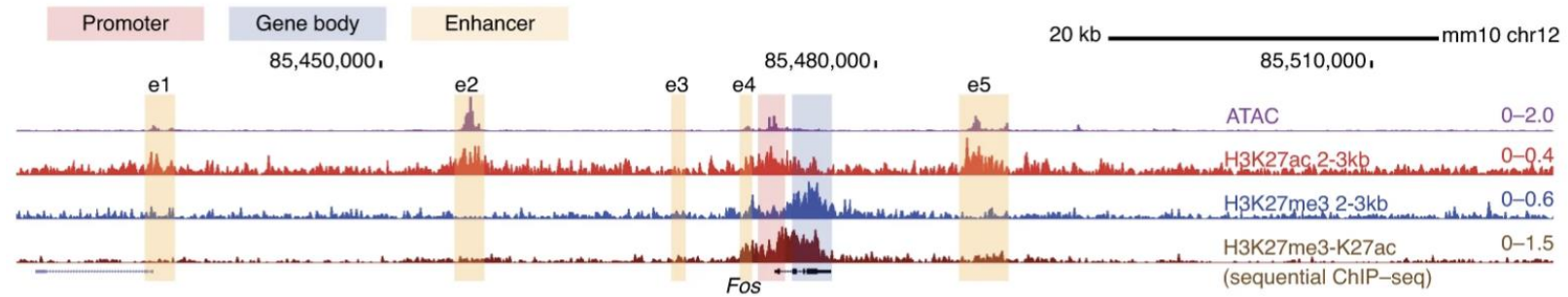
- Can target histone marks or TFs
- Input vs IP samples
 - Genome biases
- Enrichment for the target of interest, still leaves background reads
- Antibody quality
- Abundance of target
- Cell types
- Amount of starting material

Mapping the reads

- Adapter trimming (short fragments or PE data)
- Alignment to the reference genome ~ 95% alignment
- Quality control on the mapping
- Duplication levels across samples

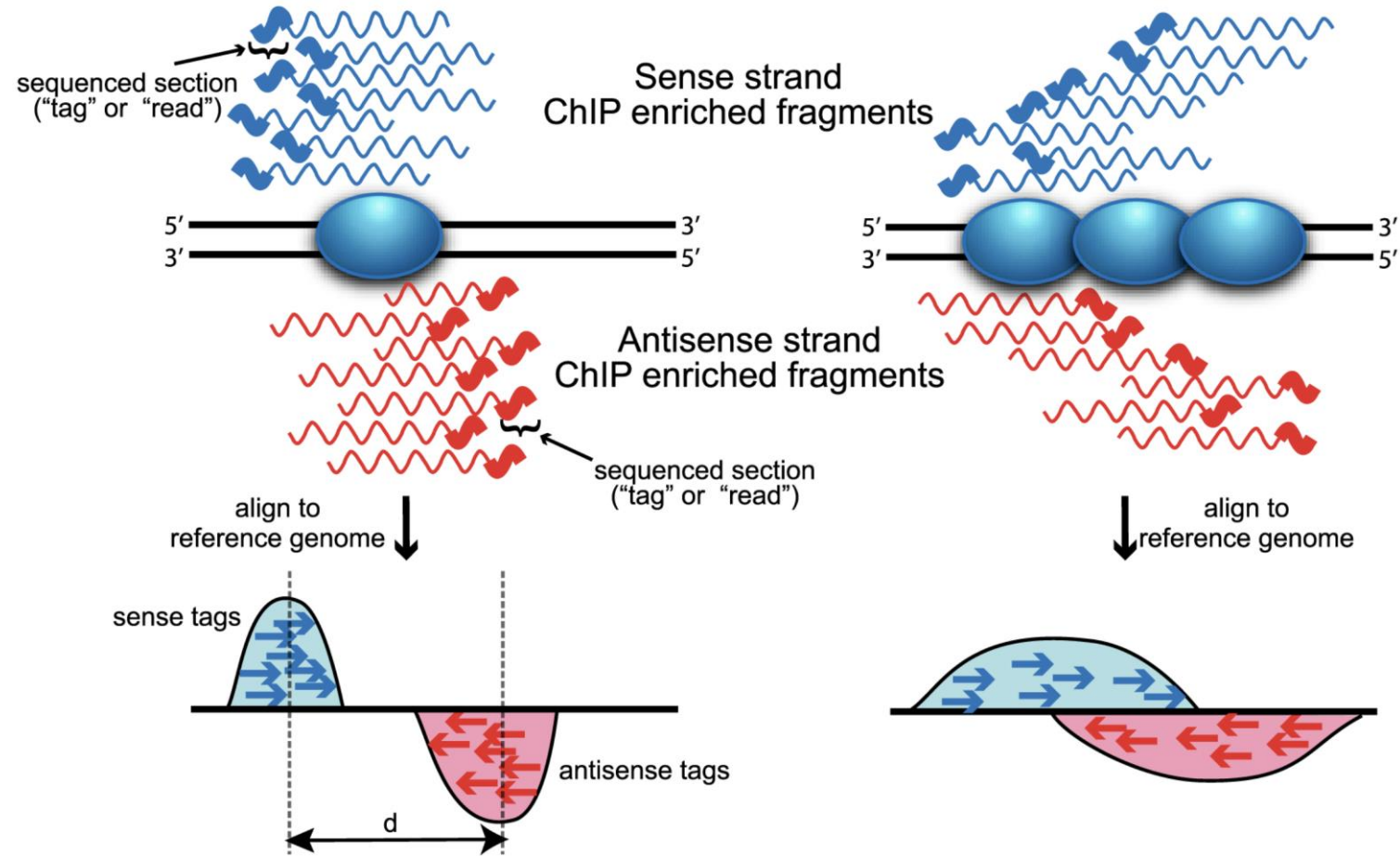
Mapping the reads

- Adapter trimming (short fragments or PE data)
- Alignment to the reference genome ~ 95% alignment
- Quality control on the mapping
- Duplication levels across samples

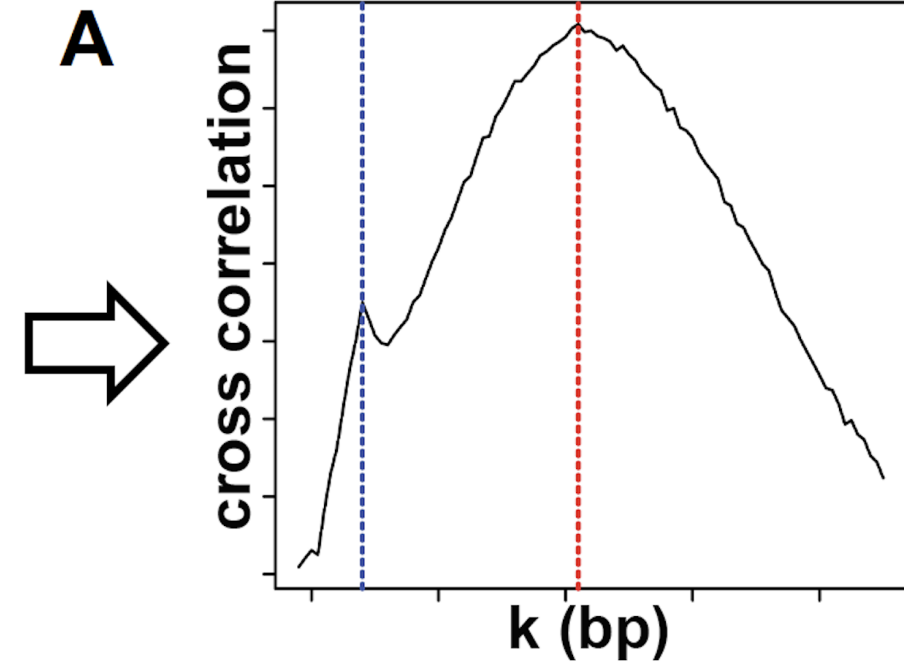
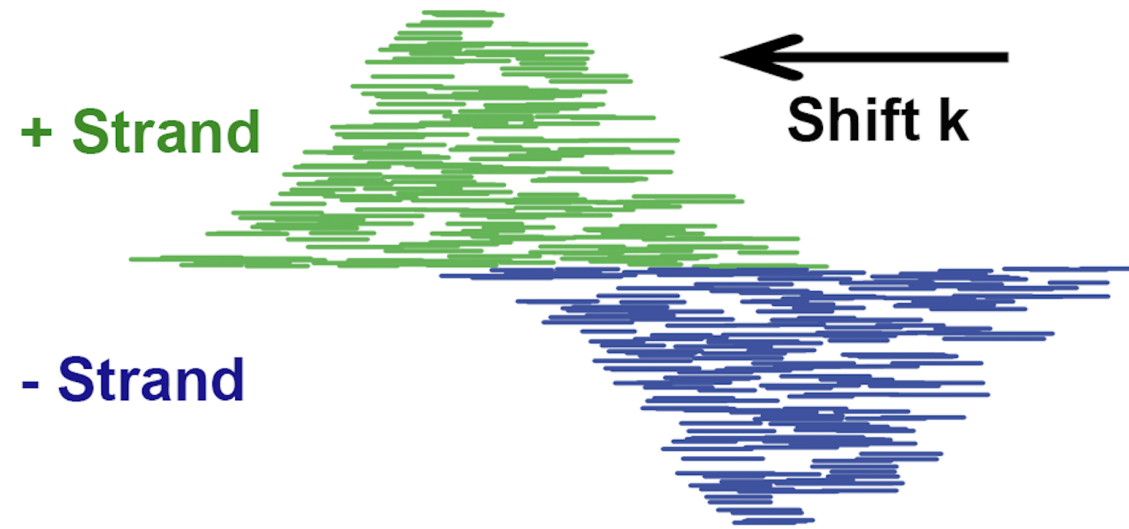


Mapping the reads

- Adapter trimming (short fragments or PE data)
- Alignment to the reference genome
- Quality control on the mapping
- Duplication levels across samples



Estimating the average fragment size

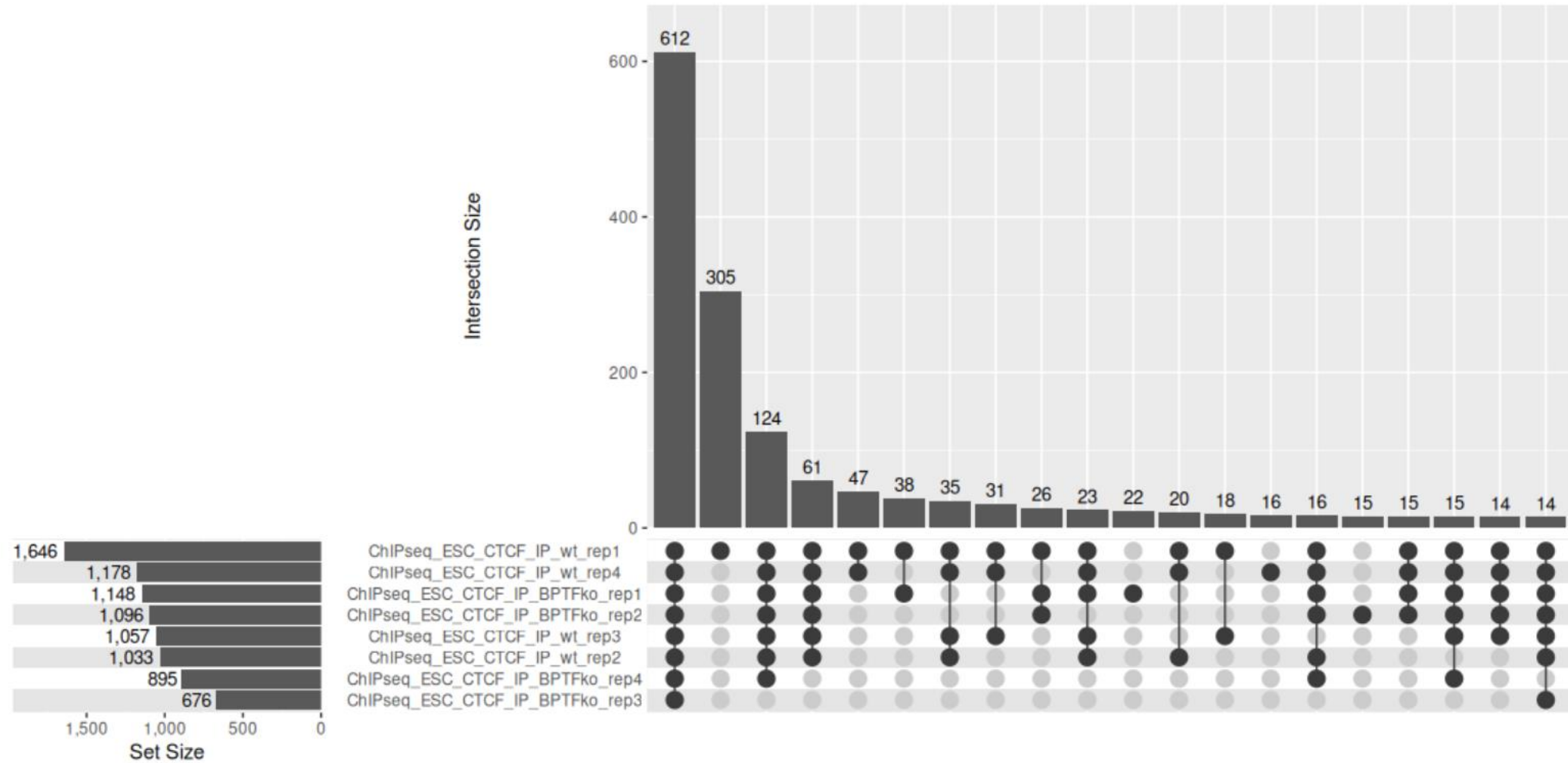


Peak calling

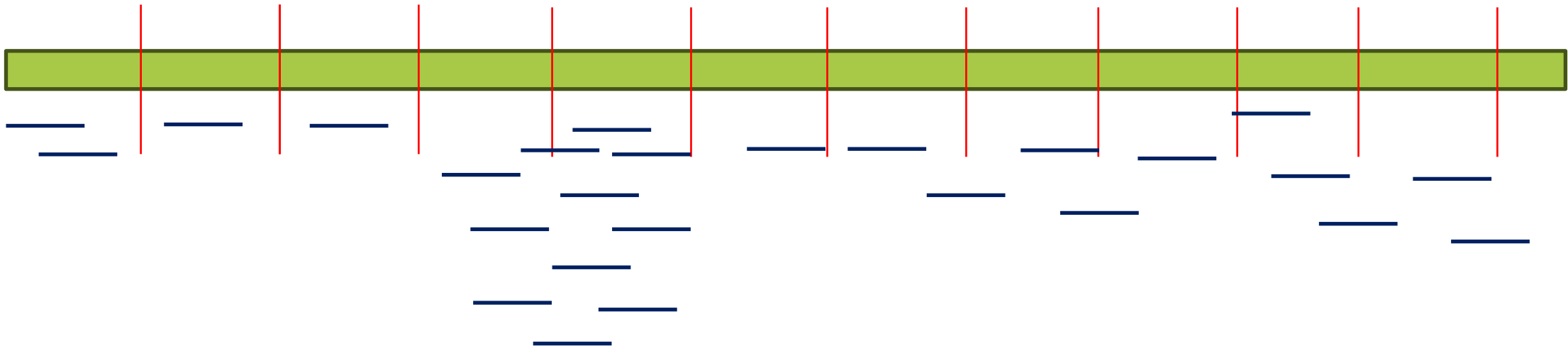
- MACS (Model-based Analysis of ChIP-seq)
- Peak caller will internally estimate the fragment size and shift the reads before peak calling
- Looking for counts above a certain background window
- Narrow peaks (TFs) vs broad peaks (histone marks)
- Example call on TF peaks (binding sites):

```
macs3 callpeak -t bamIP -c bamInput -f BAM -g genomeSize -q 0.01 \  
-n sampleName --outdir ChIPpeaks
```

Shared peaks

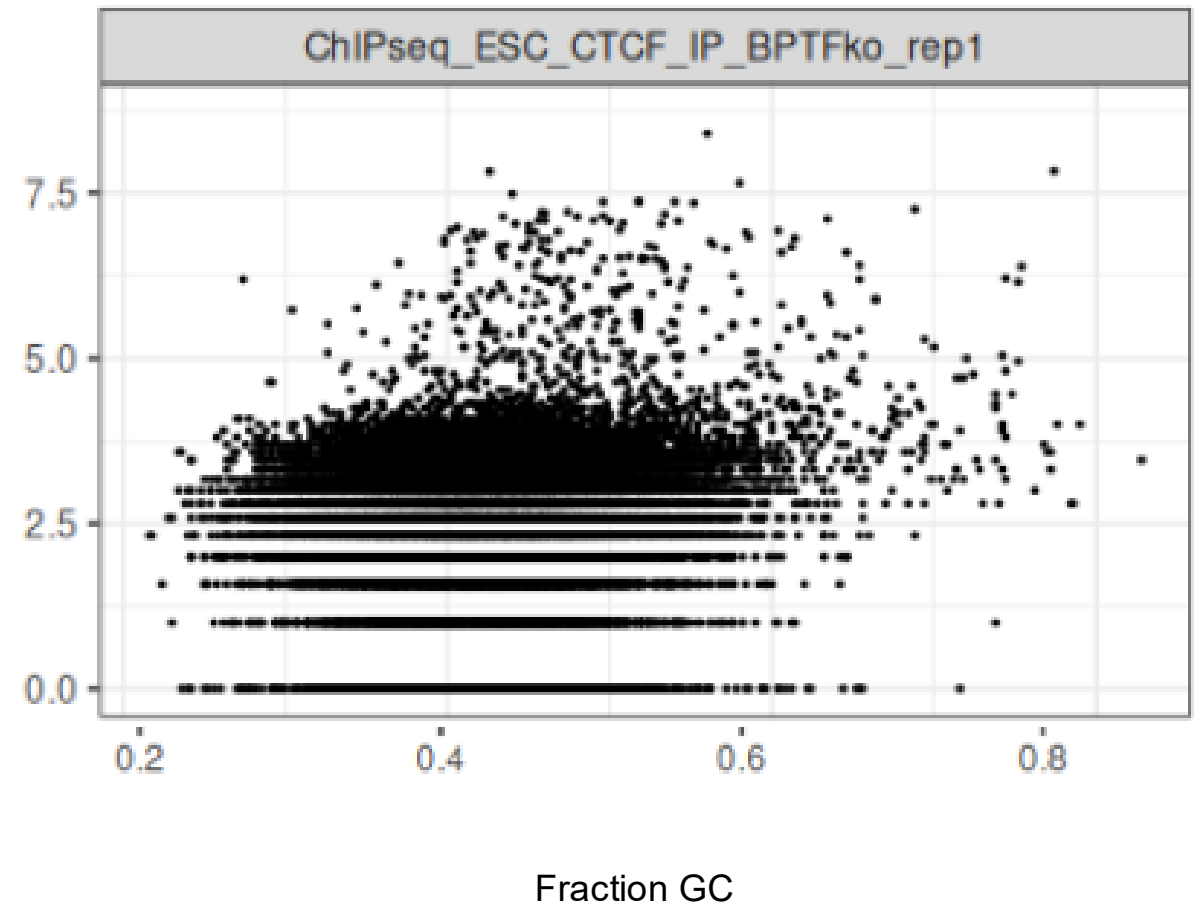
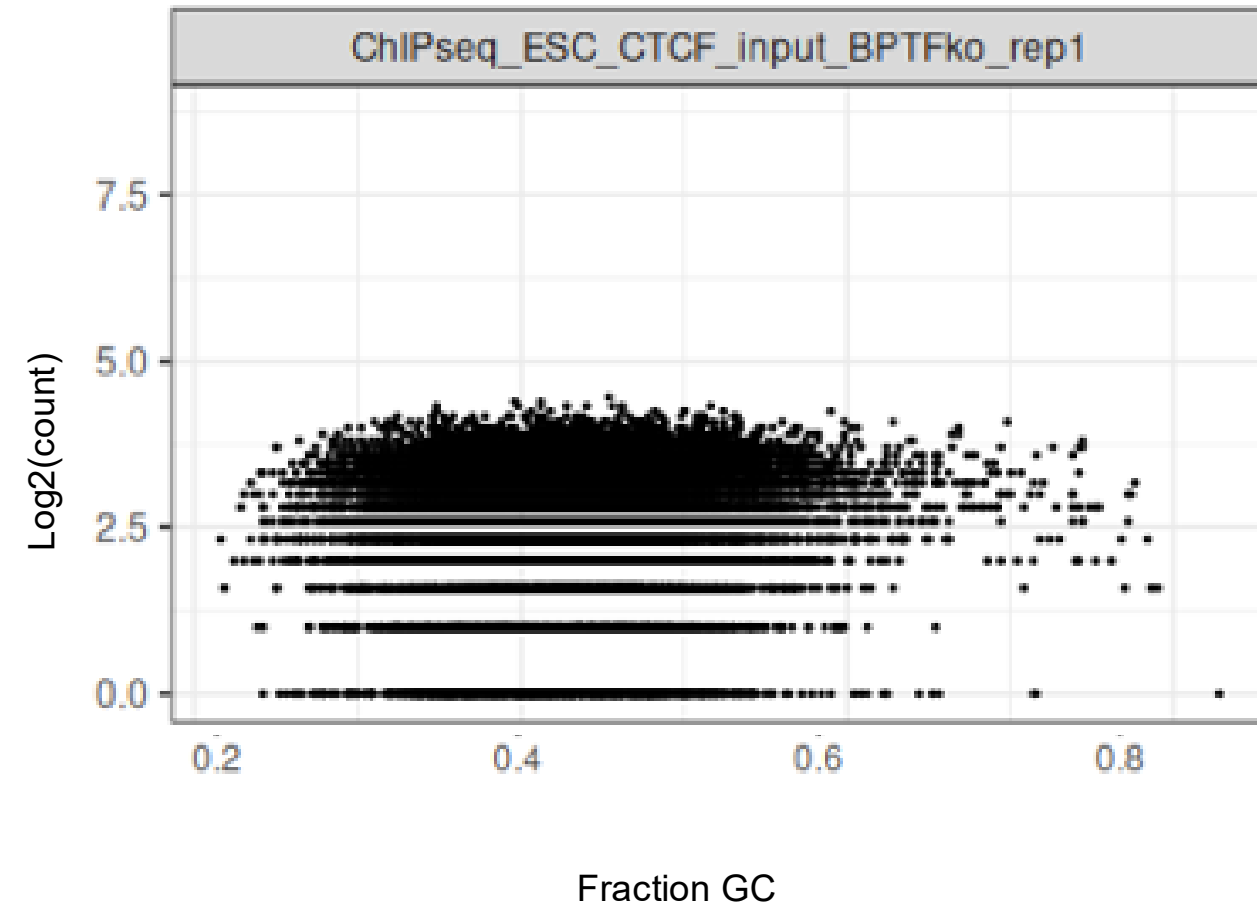


Producing tiled count matrix



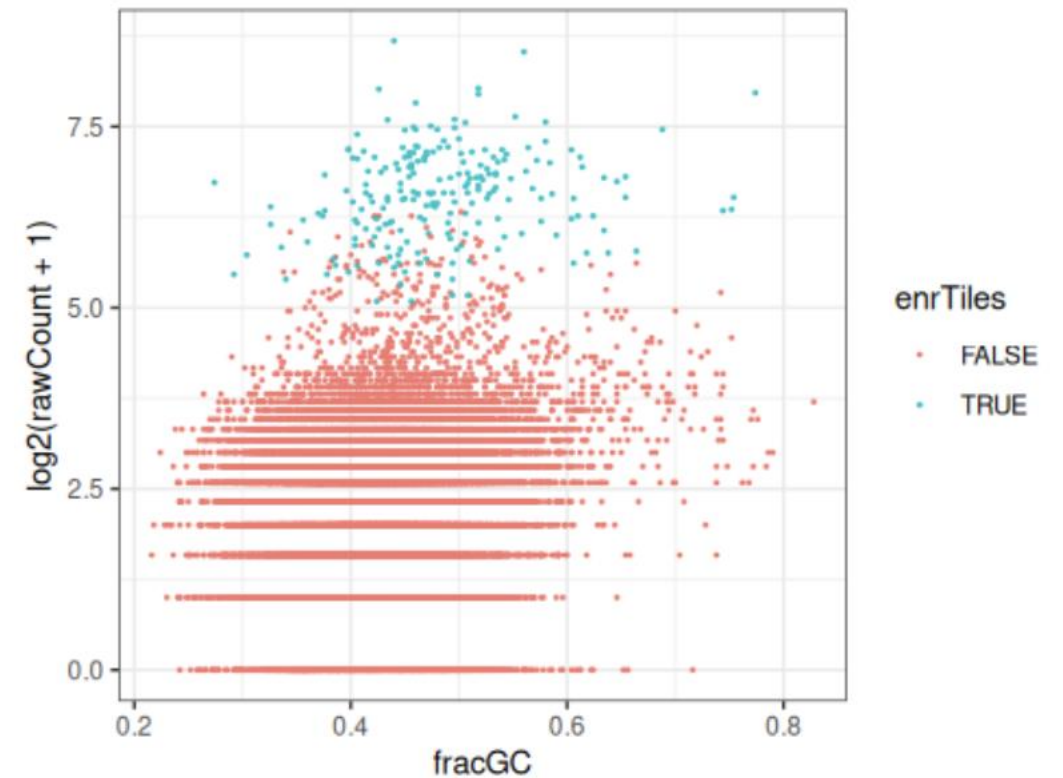
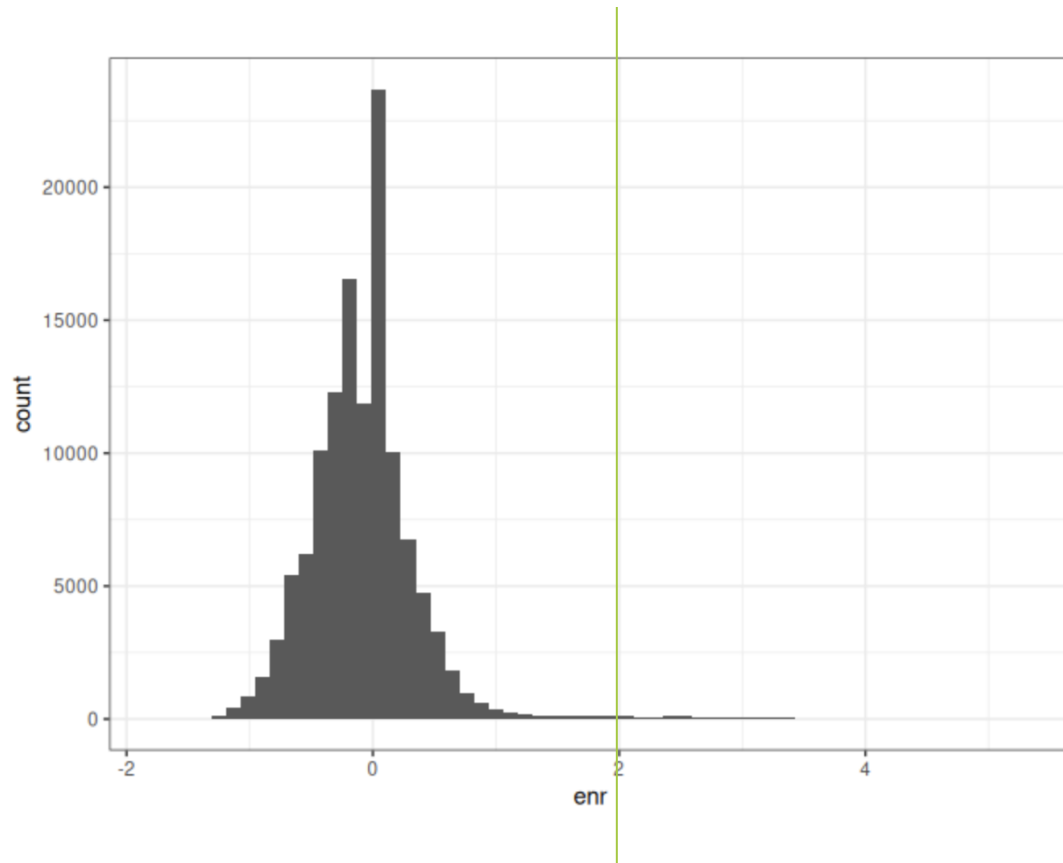
	Sample A	Sample B	fracGC
Tile_1	2	3	0.6
Tile_2	40	20	0.4
...			

Count vs GC content (no trend)

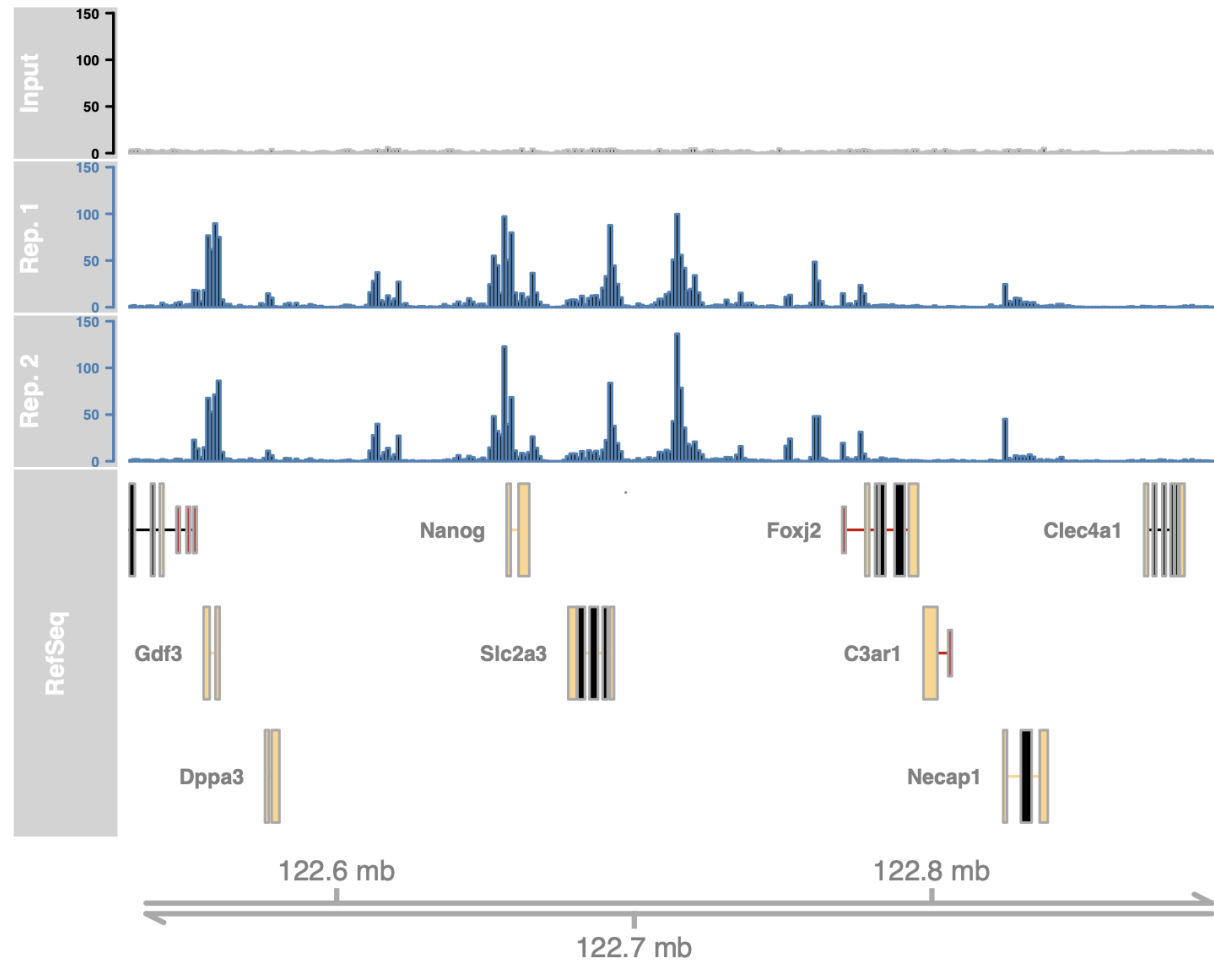


Enrichment over the input sample (TF sample)

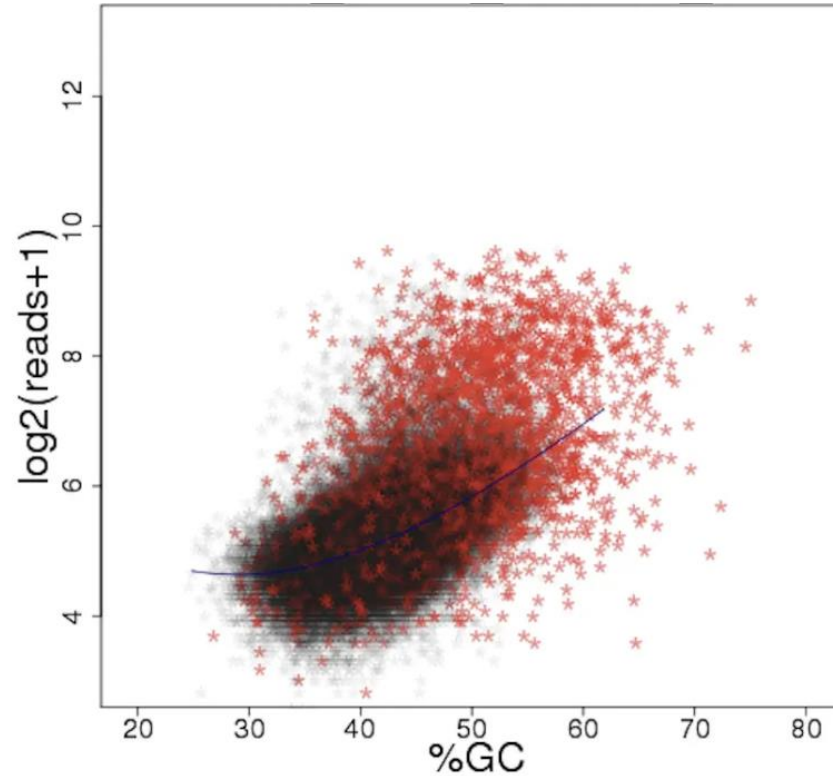
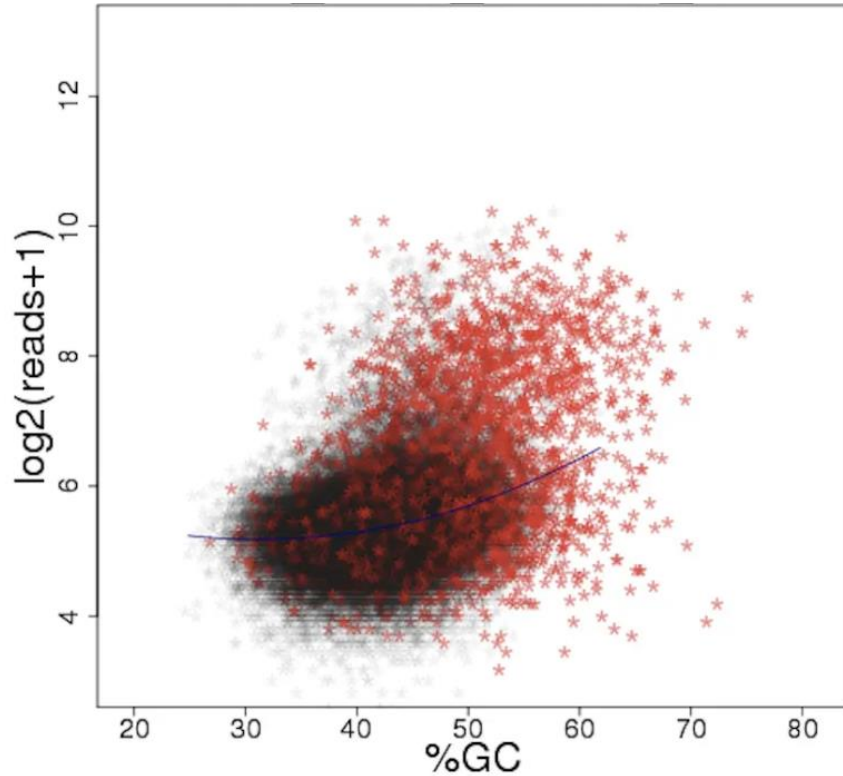
$$enr = \log_2 \frac{(normCountIP+8)}{(normCountInput+8)}$$



Genome browser views with the input

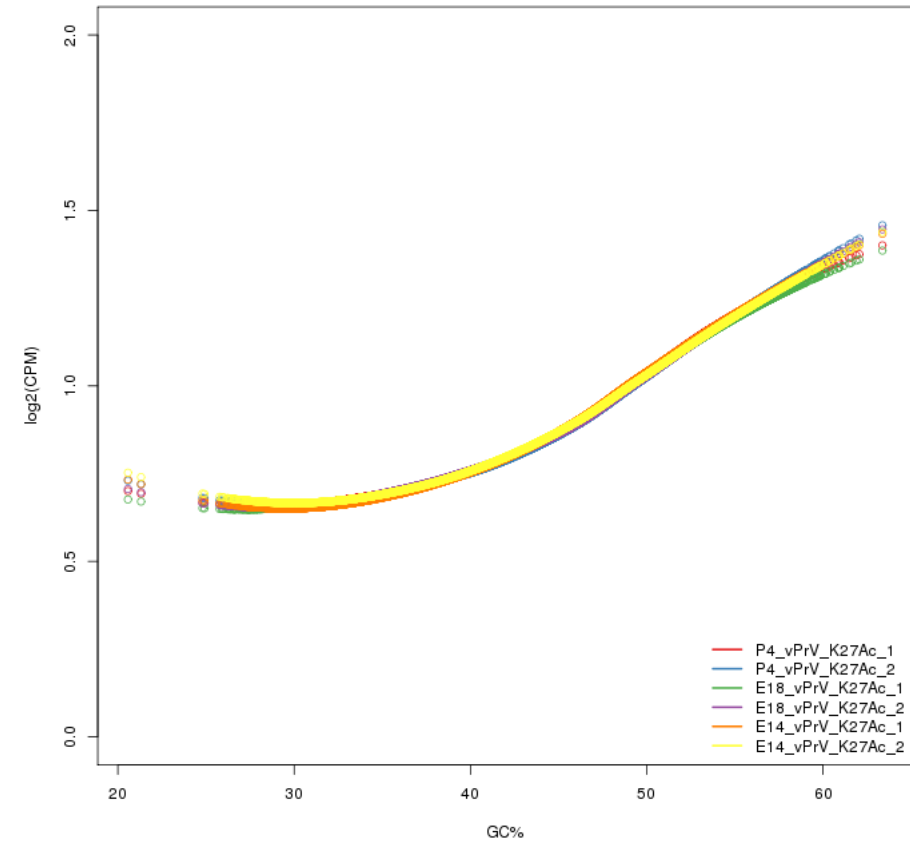
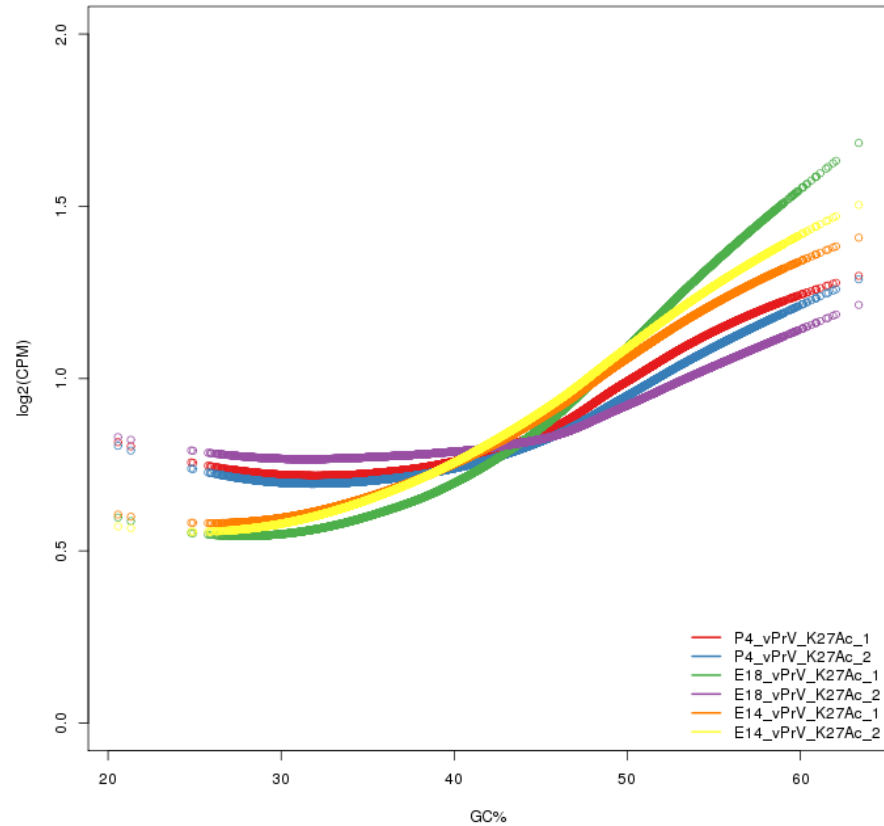


GC Bias (H3K27Ac samples)

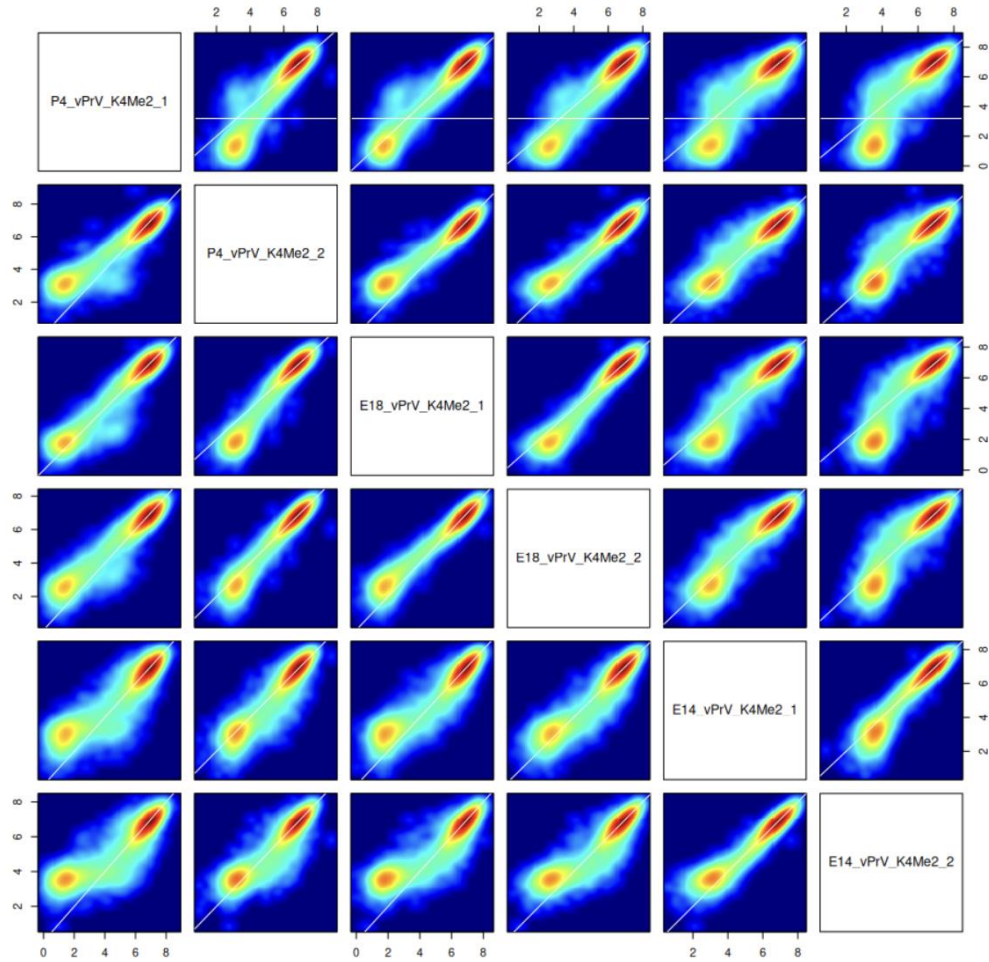


- Tiles (500bp or 5000bp)
- Global picture
- Red: CGIs
- Compare trend in background tiles
- Compare signal or enrichment over the background

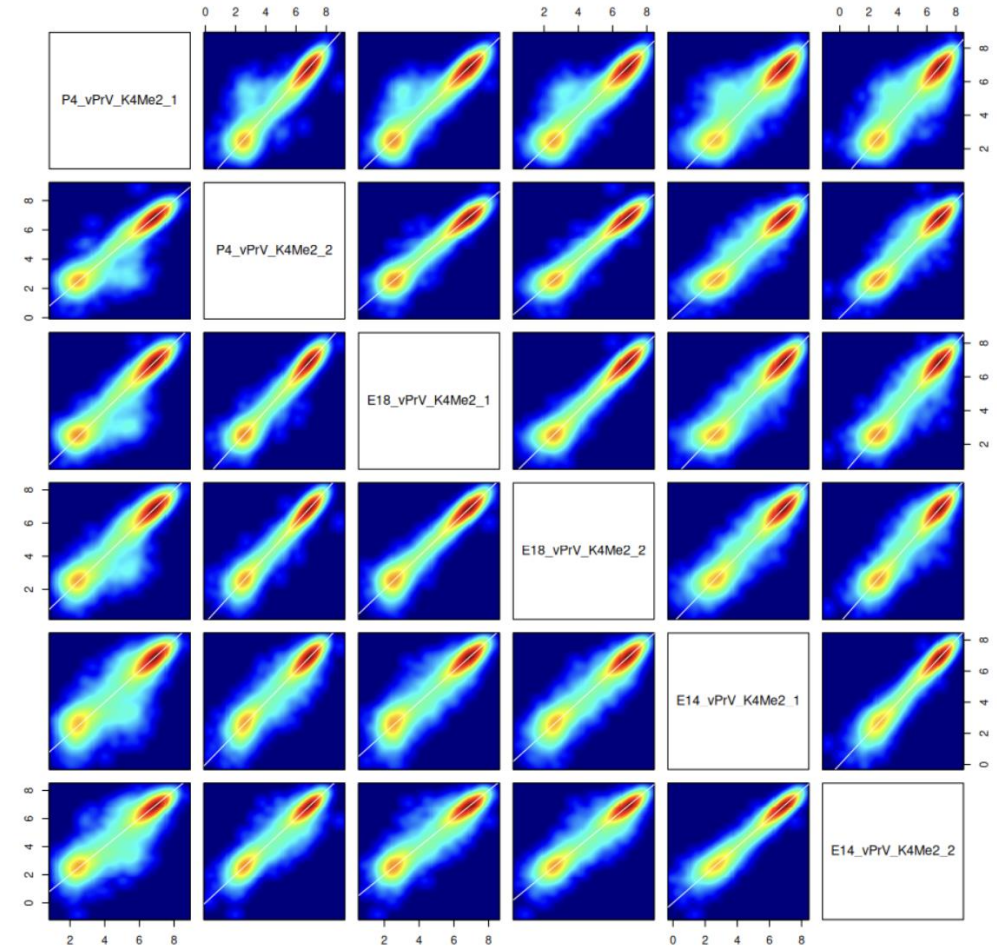
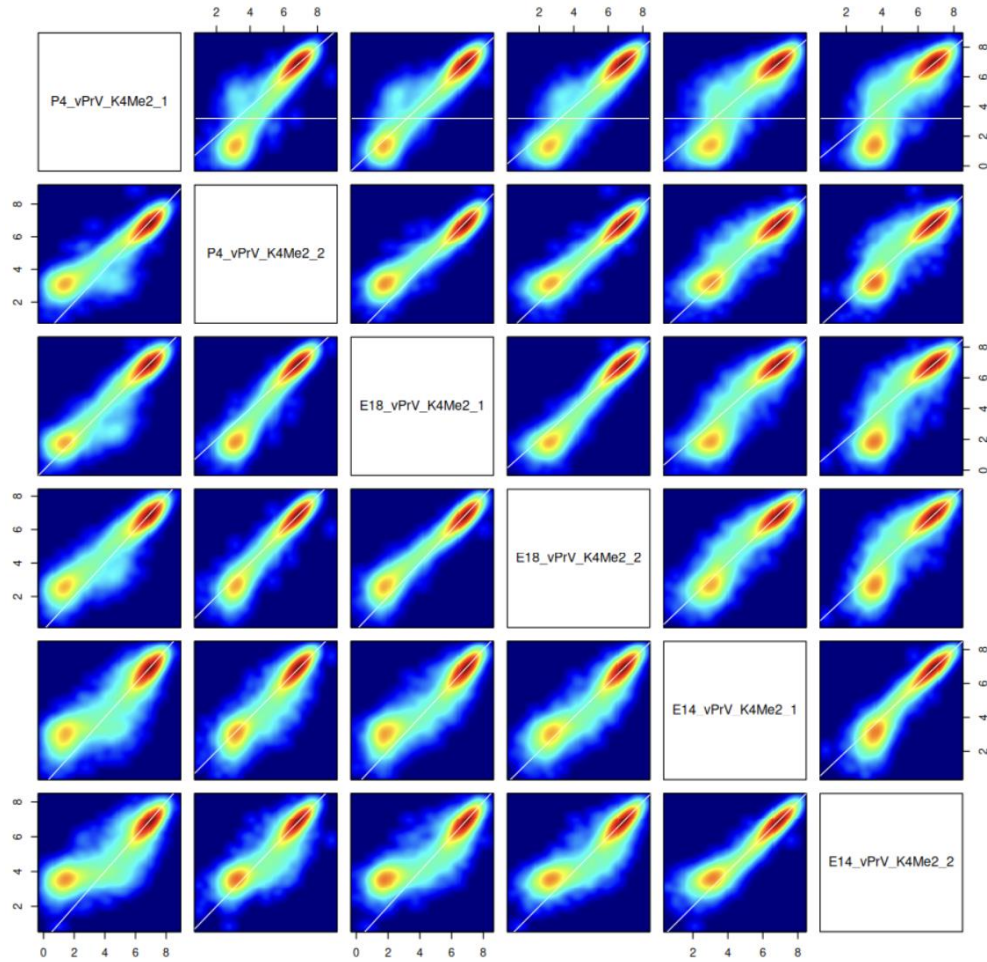
Global shifts in counts driven by differences in GC trend



Global shifts in counts driven by differences in GC trend

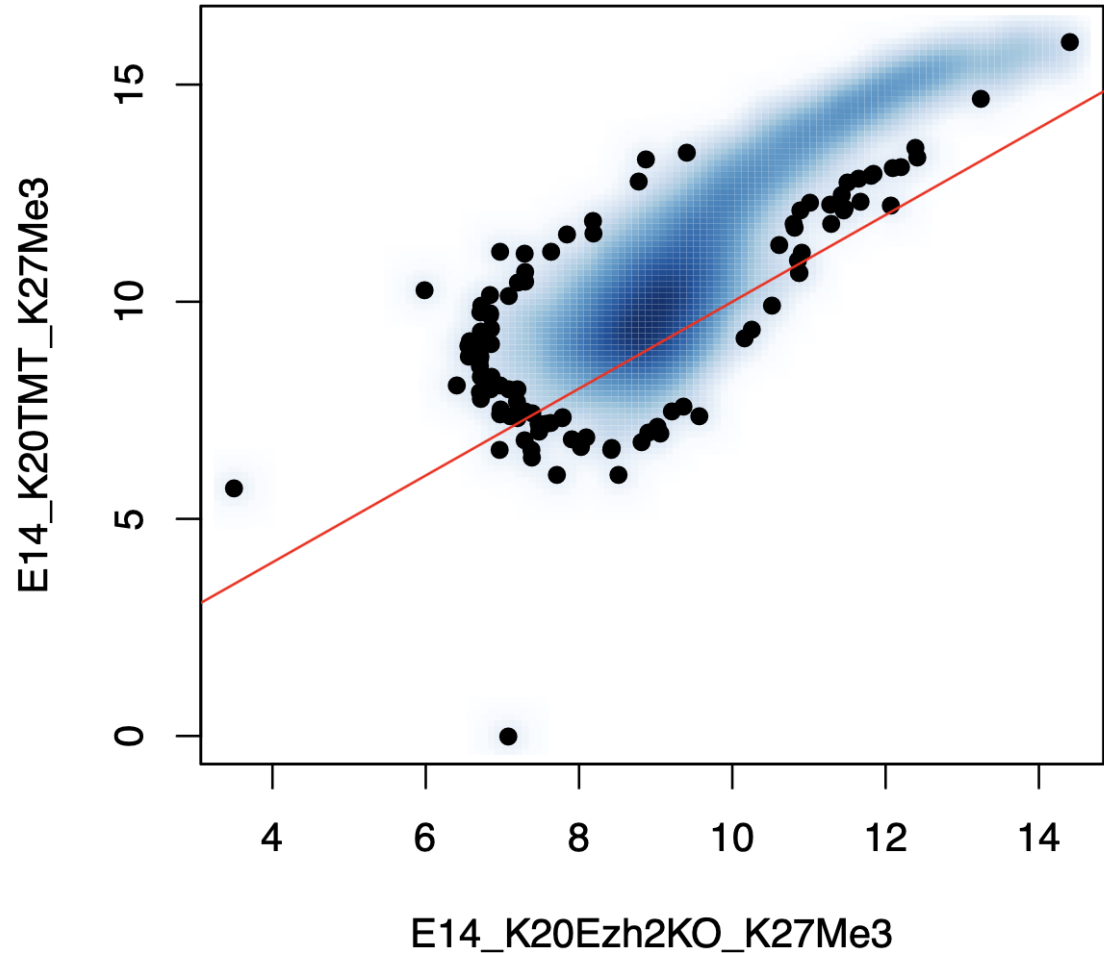


Global shifts in counts driven by differences in GC trend

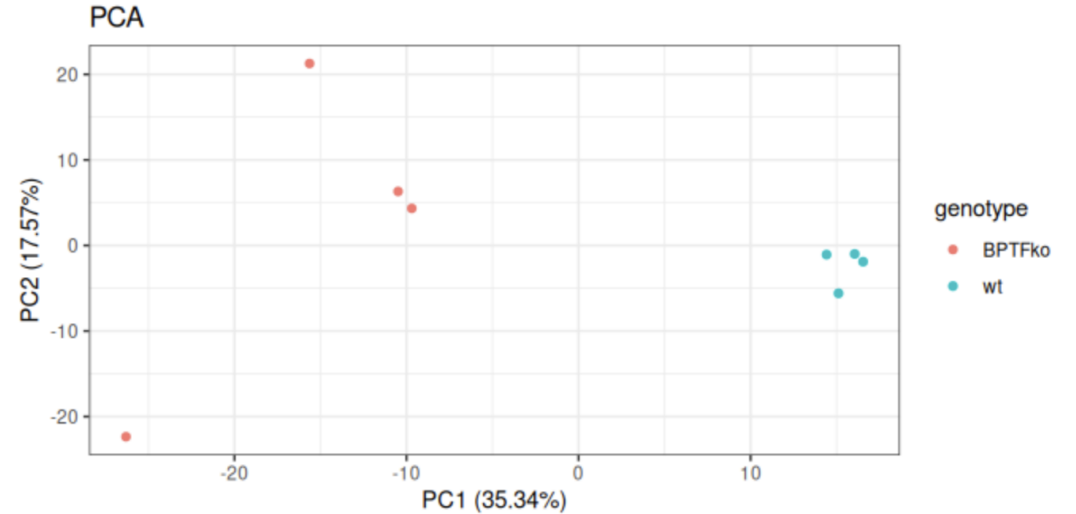
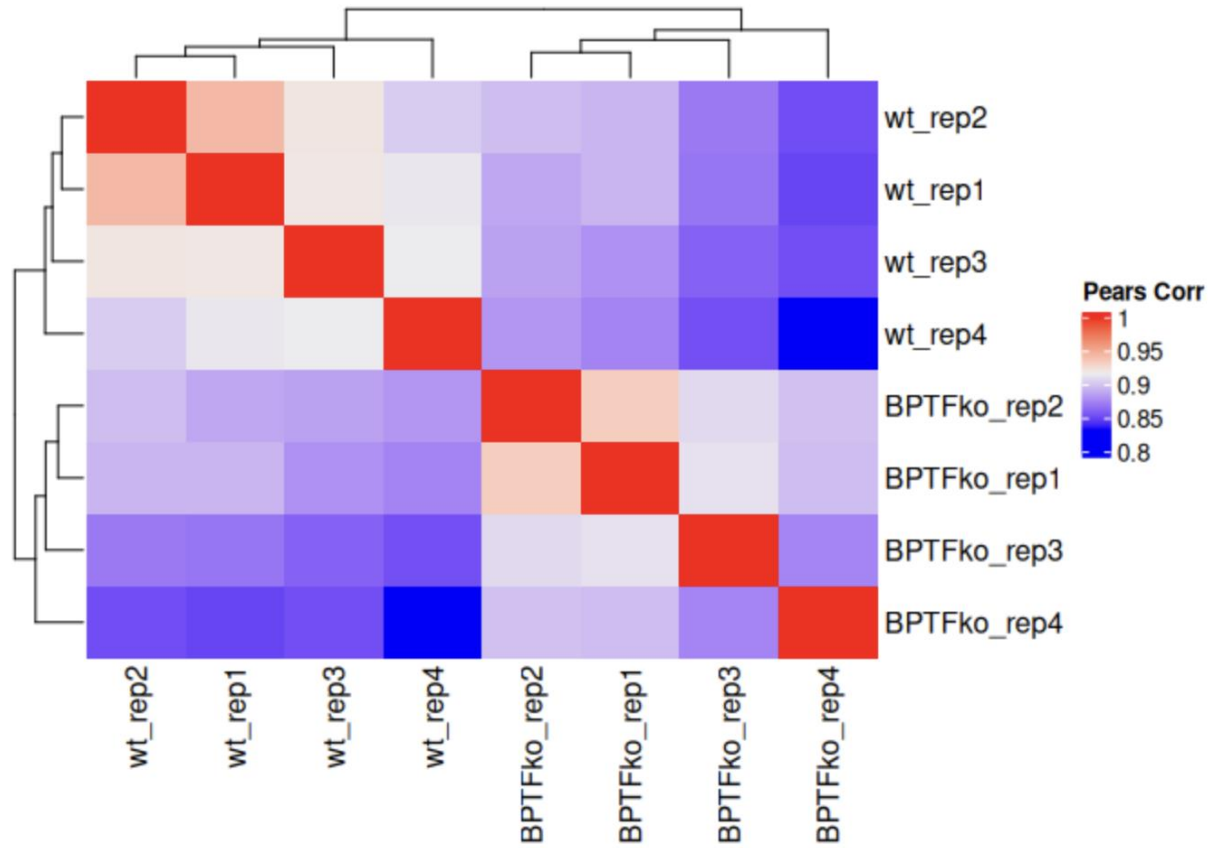


Sometimes we do expect global shifts

- Ezh2 involved in the complex mechanism with polycomb repressive complexes which adds Me3 to H3K27.
- Knocking it out would have a global impact on the H3K27Me3 mark.
- Correct normalization of the counts is tricky

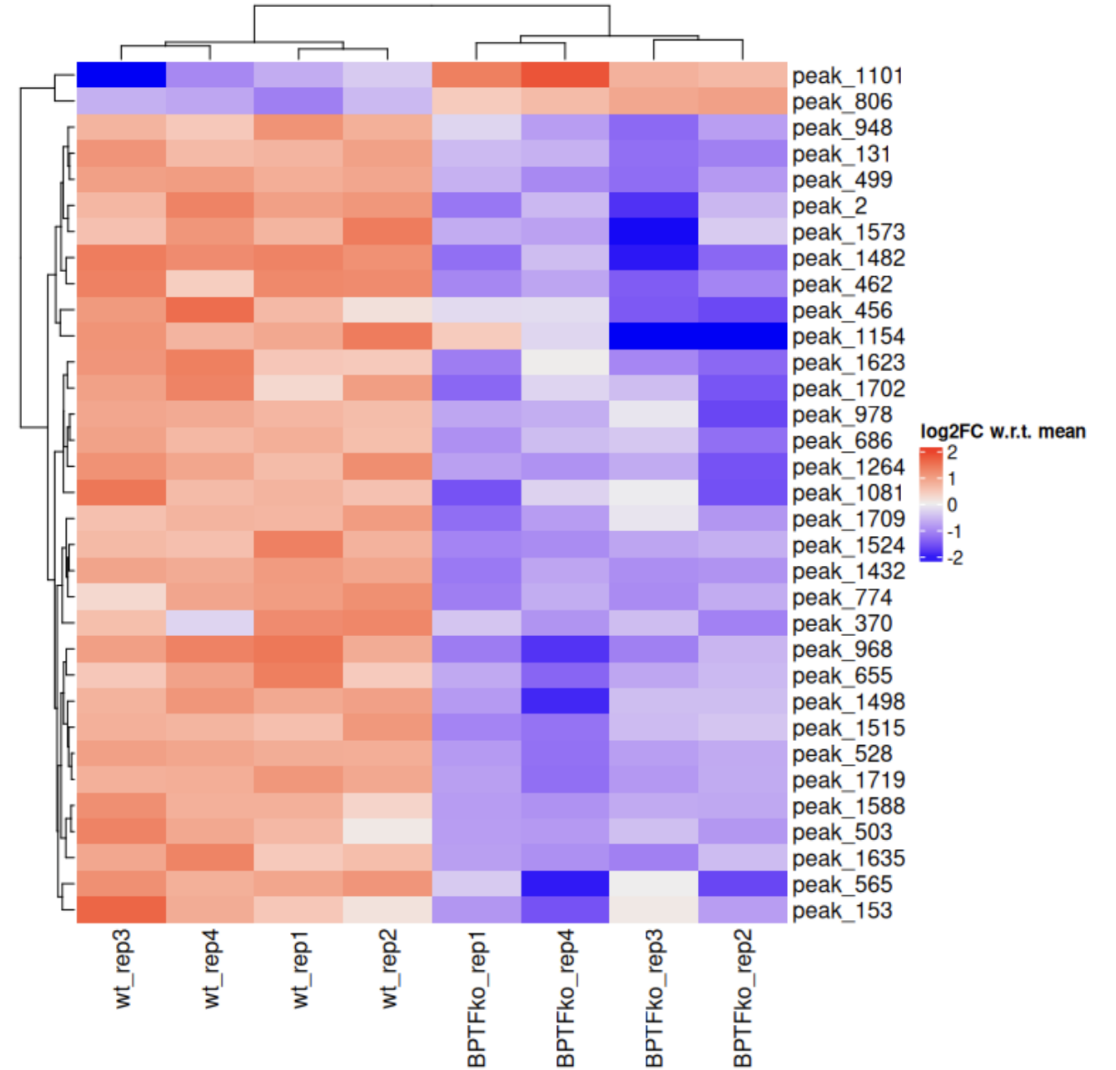
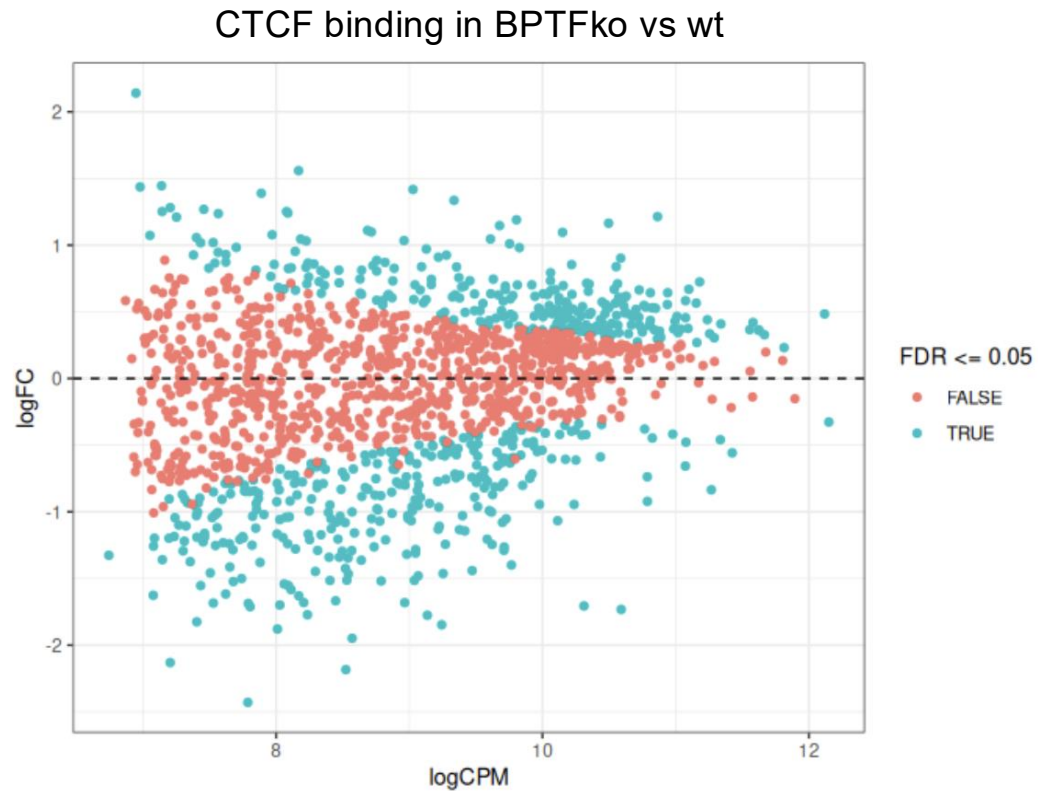


Replicates – the more the merrier

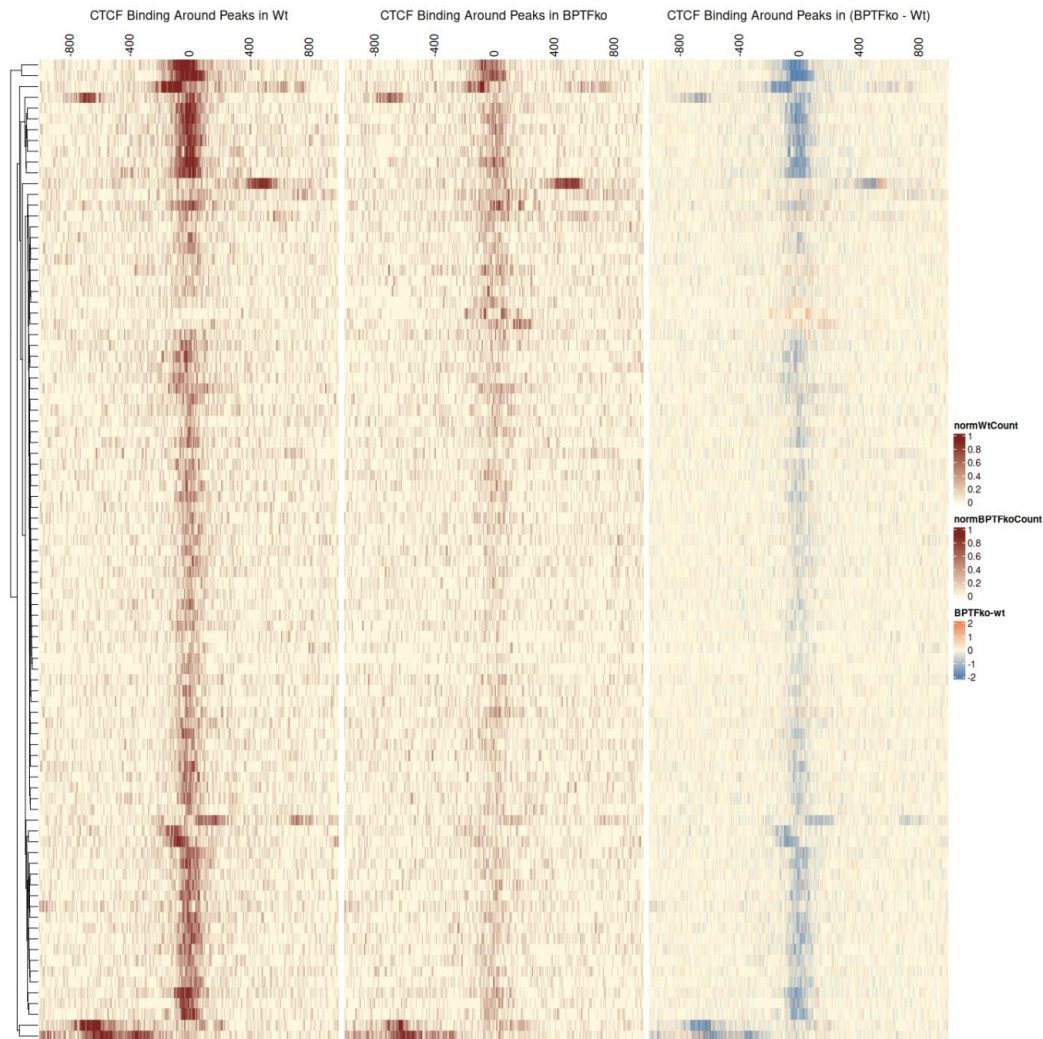


- Experiments can fail
- Biological and technical sources of variance
- Is our signal reproducible?

Do the replicates agree with differential binding?



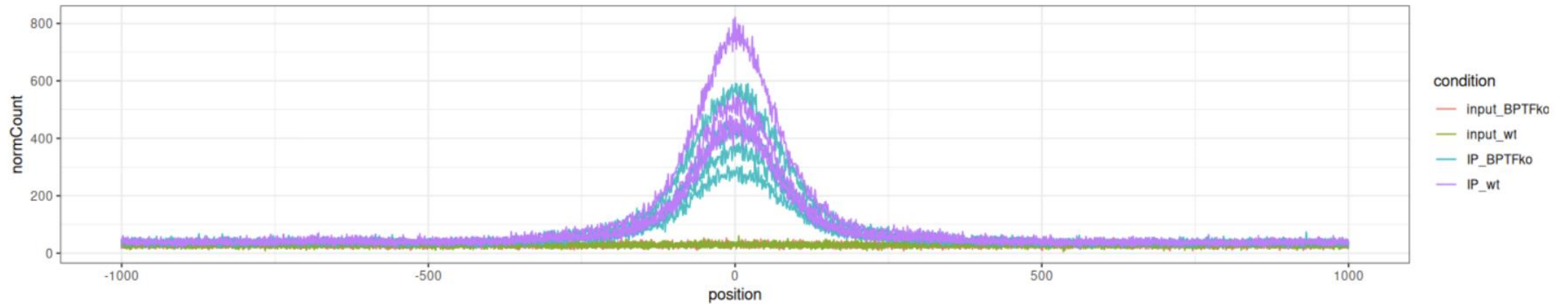
Visualizing global differences in TF binding



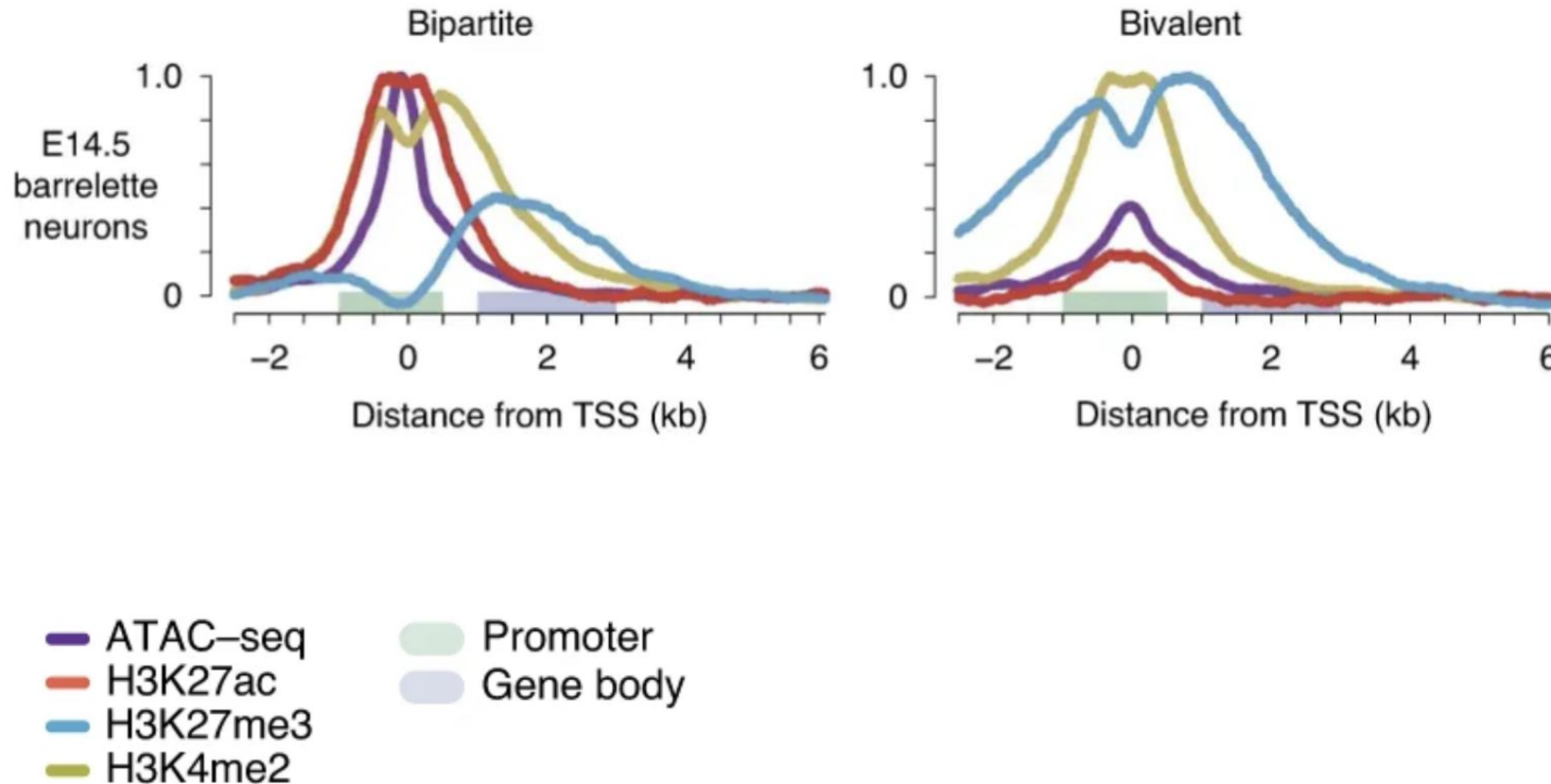
- Centered around midpoint of each peak (rows)
- Wt, BPTFko and difference

Profile plots

- Global loss in CTCF binding from wt to BPTFko



Profile plots with multiple marks

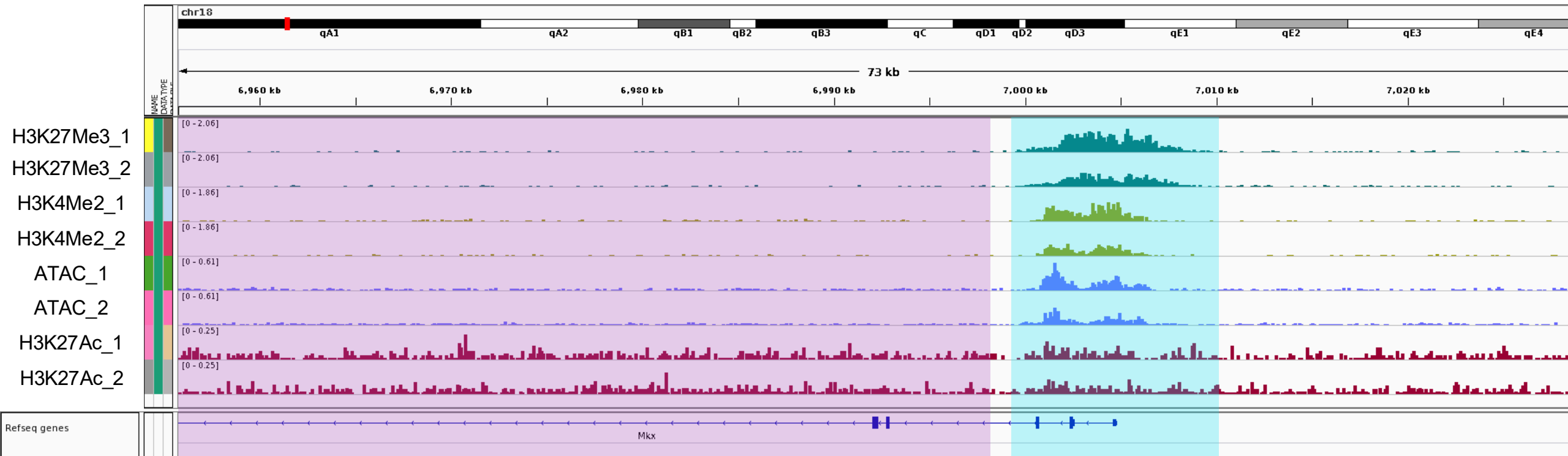


- Bipartite vs bivalent genes
- Centered around TSS
- Different epigenetic states and functional features that distinguish the two



Visualizing global changes in chromatin

- How do we visualize all genes with their corresponding chromatin profiles?



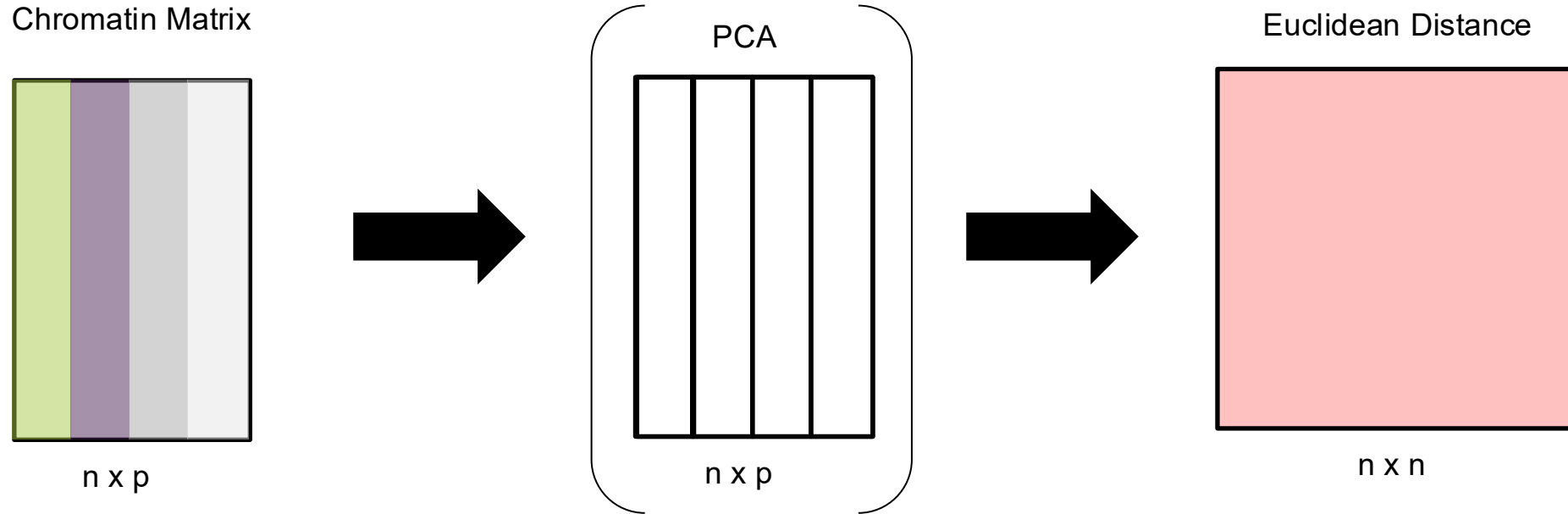
Chromatin states across genes

	<i>H3K4me2 promoter</i>	<i>H3K4me2 gene body</i>	<i>H3K27ac promoter</i>	<i>H3K27ac gene body</i>	<i>H3K27me3 promoter</i>	<i>H3K27me3 gene body</i>	<i>ATAC promoter</i>	<i>ATAC gene body</i>
<i>Gene1</i>	2.170	2.899	3.650	4.165	4.466	4.038	2.972	4.016
<i>Gene2</i>	5.687	4.895	5.614	4.563	3.436	4.101	7.188	4.576
<i>Gene3</i>	3.522	3.286	4.381	4.301	4.284	3.789	3.123	4.075
<i>Gene4</i>	2.504	2.251	3.635	4.440	3.669	3.306	2.756	3.912
<i>Gene5</i>	5.563	5.748	5.790	5.109	3.311	2.748	7.609	4.501
...								

Chromatin states across genes and time

	<i>H3K4me2 promoter</i>	<i>H3K4me2 gene body</i>	<i>H3K27ac promoter</i>	<i>H3K27ac gene body</i>	<i>H3K27me3 promoter</i>	<i>H3K27me3 gene body</i>	<i>ATAC promoter</i>	<i>ATAC gene body</i>
<i>Gene1_E14</i>	2.170	2.899	3.650	4.165	4.466	4.038	2.972	4.016
<i>Gene2_E14</i>	5.687	4.895	5.614	4.563	3.436	4.101	7.188	4.576
...								
<i>Gene1_E18</i>	2.504	2.251	3.635	4.440	3.669	3.306	2.756	3.912
<i>Gene2_E18</i>	5.563	5.748	5.790	5.109	3.311	2.748	7.609	4.501
...								
<i>Gene1_P4</i>	3.522	3.286	4.381	4.301	4.284	3.789	3.123	4.075
<i>Gene2_P4</i>	5.670	5.787	5.974	4.869	2.288	3.485	7.300	5.235

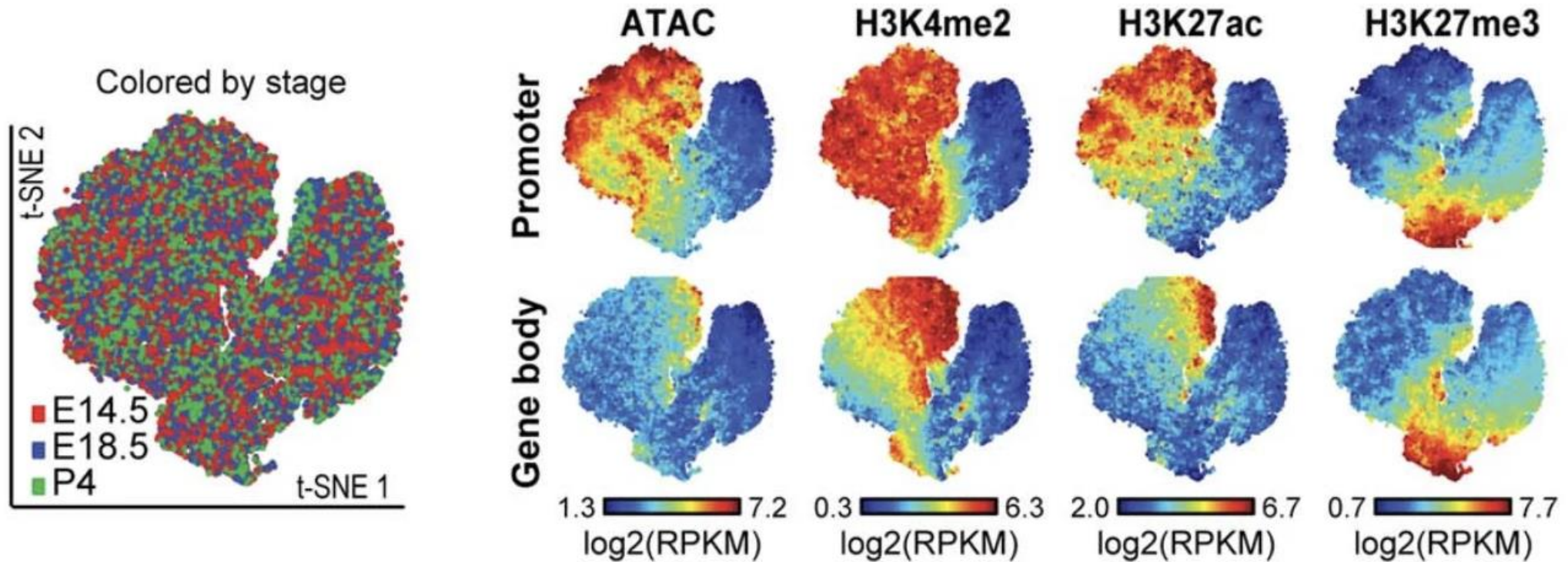
t-distributed Stochastic Neighbor Embedding (t-SNE)



- General idea with t-SNE: points which are close to each other in the high dimensional space should remain close to each other in the low dimensional space

Visualizing chromatin states globally

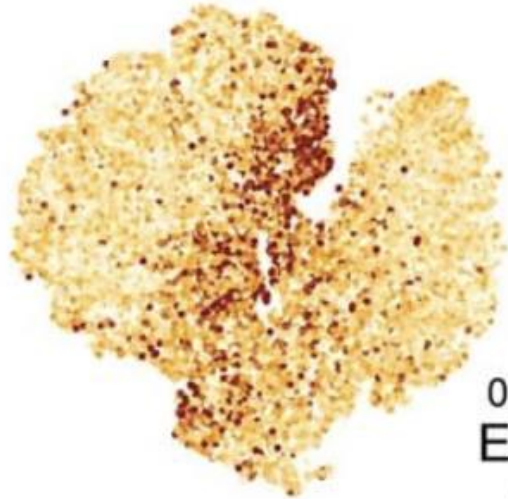
E14.5, E18.5 and P4 combined t-SNE map



Tracking chromatin changes over time

Developmental dynamics of epigenetic profiles

E14.5

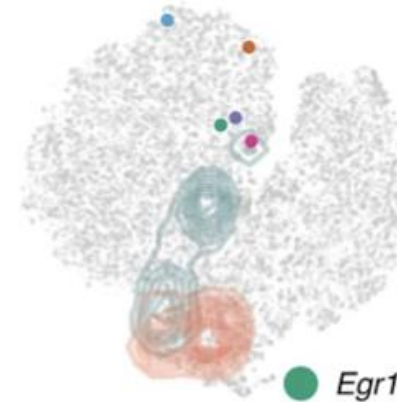


0.25  7.23
E14.5 to P4 Movement
(Euclidean Distance)

E14.5



P4



E14.5 P4
Bip → Exp

 *Egr1*  *Fos*
 *Bcl6*  *Nr3c1*
 *Plekhh3*

Tutorials

1. Introduction to ChIP-seq Analyses

- GC bias and other QC
- Estimating the fragment size
- Enrichment over input
- Peak calling
- Heatmaps of TF binding across most differential peaks between 2 conditions (wt and BPTFko)
- Profile plots
- Differential binding
- Genome tracks

2. Advanced ChIP-seq

- Correcting for non-linearities coming from different GC trends
- Visualizing the chromatin landscape with t-SNE
- Tracking changes over time

Take your time with the labs (we have it), read carefully and feel free to ask us questions and take the opportunity to discuss with your group mates to see if your solutions to the exercises agree or make sense.

Any questions?

Thank you